Oluwafemi Olabukoye

Embedded Systems & Industrial IoT Engineer

Firmware · PCB Design · Industrial Connectivity · Edge-to-Cloud Integration

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PROFESSIONAL SUMMARY

Embedded Systems and Industrial IoT Engineer with 4+ years of hands-on experience taking connected hardware from schematic to field deployment. Designs custom PCBs, writes Embedded C/C++ firmware for ESP32 and STM32, and builds Raspberry Pi/Linux edge gateways that move industrial data over Modbus RTU/RS485, CAN and MQTT to the cloud. Delivered GasFlo, an ESP32-S3 automated LPG dispensing system now running as four operational units, and currently commissions CNG metering telemetry at industrial gas facilities. Seeking remote or hybrid roles in embedded firmware, industrial IoT and edge computing.

4 units deployed
GasFlo LPG dispensers in operation

5 / 5 PCBAs verified
VESCAN board bring-up

4 gateways live
CNG gas-meter telemetry in the field

TECHNICAL SKILLS

Languages	Embedded C, C++, Python
MCUs & platforms	ESP32, ESP32-S3, ESP32-P4 (integration), STM32, ATmega, Raspberry Pi
Firmware & tools	ESP-IDF, FreeRTOS, PlatformIO, STM32CubeIDE, Embedded Linux, Git
Electronics & PCB	Schematic capture, PCB layout, prototyping, PCBA inspection, board bring-up · KiCad, Altium, EasyEDA
Industrial & automotive	Modbus RTU, RS485, UART, SPI, I2C, serial gateways · CAN, CAN-FD, K-Line, OBD diagnostics
Connectivity & cloud	Wi-Fi, BLE, LTE/GSM, GNSS, MQTT, HTTP/HTTPS, AWS IoT, OTA updates
Systems & field	Sensor integration, edge gateways, industrial telemetry, field commissioning, root-cause analysis, technical documentation, vendor & OEM coordination

PROFESSIONAL EXPERIENCE

IoT Delivery Engineer · iTech (Tetracore) · Lagos, Nigeria — Jan 2026 – Present

- Installed and commissioned 4 Raspberry Pi 4 edge gateways now active at CNG facilities, polling PTZ gas meters over Modbus RTU/RS485 in Python for remote telemetry.
- Capture pressure, temperature, measured volume (Vm), corrected volume (Vb) and interval readings for remote monitoring of industrial gas metering.
- Configure Linux services, networking and remote access on field gateways; diagnose power, meter-communication and connectivity faults on site.
- Run technical validation and hardware testing; write commissioning and handover documentation for each deployment.
- Coordinate deployment schedules and technical requirements with OEMs, vendors and client site teams.

Embedded Systems / IoT Engineer · Zubi Technologies — 2022 – 2025

- Delivered embedded hardware and firmware for two vehicle products: **VESCAN**, an automotive diagnostics and telematics device (CAN/CAN-FD, K-Line, cellular), and a vehicle safety and emergency-communication system (STM32, AWS IoT) taken to pilot.
- Brought up and tested VESCAN PCBAs (all five received boards verified working) and coordinated fabrication with PCB vendors.
- Developed Embedded C/C++ firmware and hardware communication interfaces; designed circuit prototypes and contributed to schematic capture and PCB development.
- Integrated, tested and debugged prototype hardware and firmware to find and fix system-level faults.

Software Tester & Product Manager · MedTech Africa — 2021

- Tested and quality-assured health-technology products; wrote product requirements and coordinated delivery between stakeholders and developers.

KEY PROJECTS

GasFlo — Automated LPG Dispensing System 4 units deployed

ESP32-S3 · custom PCB · HX711 load cells · ultrasonic sensing · SSR · ESP32-P4 touchscreen · BLE

- Designed and tested the prototype and custom PCB; wrote the ESP32-S3 control firmware for a token-based LPG dispenser.
- Built load-cell weighing, ultrasonic cylinder detection and solid-state-relay dispensing that stops at the purchased quantity.
- Integrated backend token validation, a live-progress ESP32-P4 touchscreen and BLE thermal receipt printing.
- Validated the end-to-end workflow and helped deploy four operational units.

VESCAN — Automotive Diagnostics & Telematics · Zubi Technologies **Prototype → production readiness**

ESP32 · CAN/CAN-FD · K-Line · cellular · GNSS · custom PCB

- Contributed to the embedded hardware architecture and custom PCB for a vehicle diagnostics and telematics device.
- Supported CAN/CAN-FD and K-Line interface work and cellular telemetry integration.
- Inspected, brought up and tested PCBAs: all five received boards verified working.
- Coordinated PCB fabrication with vendors; supported enclosure development and CE testing requirements.

Vehicle Safety & Emergency Communication System · Zubi Technologies **Pilot**

STM32 · motion sensing · MQTT · AWS IoT · OTA · SMS fallback

- Supported firmware testing, OTA updates and SMS-fallback emergency communication on STM32-based vehicle safety electronics.
- Designed and tested vehicle audio relay interface boards; supported pilot hardware integration.

Additional work: ESP32-S3/HX711 oil-drum weighing application; industrial dispenser integration against OEM API and telemetry requirements; RS485 gateway and connectivity prototypes.

EDUCATION

B.Eng. Computer Engineering — Second Class Upper (2:1) · University of Ilorin, Nigeria 2022

National Diploma, Science Laboratory Technology — Distinction · Kwara State Polytechnic, Nigeria 2015

Professional development: AWS IoT Core, embedded Linux, industrial networking, IoT system architecture.